CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 10: Arithmetic Progression

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Sum of n terms

1. An A.P. has first term 5 and last term 50, and its terms add up to 440. Find (a) the number of terms (b) the common difference. ICSE 2025, 3 marks

2. The nth term of an A.P. is $T_n = 6(7 - n)$. Find (a) its first term and common difference (b) the sum of its first 25 terms. ICSE 2024, 3 marks

3. The 5th term of an A.P. is 4 and its 9th term is -12. Find (a) the first term (b) the common difference (c) the sum of the first 16 terms. ICSE 2023, 3 marks

4. Which term of the A.P. $15, 30, 45, 60, \ldots$ is 300? Find the sum of all the terms up to it. ICSE 2023, 3 marks

5. The 6th term of an A.P. is four times its first term, and the sum of its first six terms is 75. Find the first term and the common difference. ICSE 2020, 3 marks

6. The 4th and 6th terms of an A.P. are 8 and 14. Find (a) the first term (b) the common difference (c) the sum of the first 20 terms. ICSE 2019, 4 marks

7. The 4th term of an A.P. is 22 and the 15th term is 66. Find the first term and the common difference, and then the sum of the first 8 terms. ICSE 2018, 4 marks

2 The general term

8. Find the 7th term of the A.P. $\frac{1}{a}, \frac{1}{a} + 1, \frac{1}{a} + 2, \ldots$ (a) $\frac{1}{a} + 6$ (b) $\frac{1}{a} + 7$ (c) $\frac{1}{a} + 8$ (d) $\frac{1}{a} + 7^7$ ICSE 2024, 1 mark

9. The nth term of an A.P. is $2n + 5$. Find its 10th term. (a) 7 (b) 15 (c) 25 (d) 45 ICSE 2023, 1 mark

10. $(k - 3), (2k + 1)$ and $(4k + 3)$ are three consecutive terms of an A.P. Find k. ICSE 2018, 3 marks

3 Choosing terms conveniently

11. The first three terms of an A.P. add up to 42, and the product of the first and third terms is 52. Find the first term and the common difference. ICSE 2019, 3 marks

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